

SADHVIK CHIRUNOMULA

Senior DevOps & Platform Engineer · Cloud-Native Infrastructure · Multi-Cloud (GCP, Azure, AWS)

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PROFESSIONAL SUMMARY

Senior DevOps & Platform Engineer with **6.5 years** of hands-on experience architecting, automating, and scaling multi-tenant cloud-native data platforms across **GCP, Azure, and AWS**. Currently leading a **30-engineer cross-functional team** (all direct reports) delivering a **SaaS data-engineering platform** used by **500+ data engineers** at Fortune 500 healthcare and life-sciences enterprises. Deep expertise in **Terraform IaC, Kubernetes orchestration (GKE/EKS/AKS), CI/CD automation, Apache Spark infrastructure, cost-optimised compute scheduling**, and production observability at scale. Track record of saving up to **\$70K/month** in infrastructure costs, reducing MTTR by **40%**, and cutting deployment errors by **70%**. Proven experience in **pre-sales engineering, product demos, and enterprise POC delivery** at industry summits including the Data Engineering Summit (DES). Actively leverages **AI-assisted development** (GitHub Copilot, Antigravity, Claude Code, GPT Codex) to accelerate IaC authoring, pipeline development, and operational automation.

TECHNOLOGIES

Languages	Java, Python, Dart, Bash, HCL (Terraform), SQL, YAML
Frameworks	Spring Boot, Flask, Flutter, Apache Spark (PySpark, Spark SQL)
IaC & Containers	Terraform (modules, workspaces, remote state), Docker (multi-stage builds, Compose), Kubernetes (Helm, Kustomize, RBAC, HPA, PDB, Network Policies), LXC, Firecracker microVMs, Vagrant, Packer
CI/CD & GitOps	GitHub Actions, Azure DevOps Pipelines, Ansible, Puppet, Chef, Salt, Git (branching strategies, mono-repo), PyPI, Maven Central, Semantic Versioning
Testing	Pytest, Selenium, OWASP ZAP, Integration/E2E test automation, UI regression testing
Cloud — GCP	GKE, Cloud Run, Compute Engine, Cloud Functions, Pub/Sub, BigQuery, Dataproc, GCS, Cloud SQL, VPC, IAM, Cloud Build, Stackdriver, Cloud Armor
Cloud — Azure	AKS, Azure VMs, App Service, ADLS Gen2, Azure SQL, Azure DevOps, ARM/Bicep, Azure AD (SAML/OAuth), Azure Monitor, NSGs
Cloud — AWS	EKS, EC2, S3, RDS, Lambda, EMR, CloudFormation, IAM, CloudWatch, Route 53, ALB/NLB, AWS Marketplace
Observability	Grafana, Prometheus, Alertmanager, Splunk, Dynatrace, EFK/ELK Stack, Telegraf, InfluxDB, Stackdriver, Apache Spark History Server
Data & Messaging	Apache Kafka, Confluent Kafka (Schema Registry, Connect), RabbitMQ, StreamSets Data Collector, Apache Hive Metastore, Delta Lake, Apache Iceberg
Databases	PostgreSQL, MySQL, Oracle DB, MongoDB Atlas (sharding, replica sets), InfluxDB, Cloud SQL, Azure SQL
Security	JFrog Xray, Snyk, Trivy, SonarQube, Azure AD SSO (SAML 2.0, OAuth 2.0), Ingress/Egress policies, Firewall rules, Security Groups, Secrets Management (Vault, GCP Secret Manager)
Networking	VPC (peering, shared VPC, private Google access), Load Balancers (ALB, NLB, GCP LB), DNS (Route 53, Cloud DNS), Nginx, Traefik, Istio service mesh, HAProxy
AI & Tooling	GPT-4, Gemini, Claude · GitHub Copilot, Antigravity, Claude Code, GPT Codex · LLM-assisted IaC generation, pipeline debugging, incident triage

EXPERIENCE

Lead Product Engineer | Modak Analytics, Hyderabad

Sep 2024 – Present

Leading engineering for a multi-cloud SaaS data-engineering platform serving Fortune 500 healthcare & life-sciences enterprises. Platform enables cost-optimised Apache Spark workloads across GCP, Azure, and AWS with a custom control plane, CLI, REST API, and web UI.

- Led **30 direct reports** across UI, API, QA, Database, and Platform tracks — owned hiring pipelines, mentorship, sprint planning, architecture reviews, and cross-track delivery accountability with a **zero-attrition record** in the first year.
- Drove **product roadmap and feature prioritisation** directly with C-suite and enterprise stakeholders, translating business goals into quarterly milestones and achieving **100% on-time delivery** across 4 consecutive quarters.
- Architected the **multi-tenant SaaS control plane** that orchestrates Apache Spark cluster lifecycles (provisioning, auto-scaling, auto-stop, teardown) across GCP Dataproc, AWS EMR, and Azure HDInsight, serving **500+ data engineers** running **10K+ Spark jobs/week**.
- Designed a **custom compute scheduler** for dynamic VM orchestration, implementing intelligent auto-scaling with preemptible/spot instance failover, checkpointing, and graceful migration — achieving **60–80% compute cost reduction** vs. on-demand pricing.
- Built the platform **CLI in Python**, packaged and **published to PyPI** with semantic versioning, automated release pipelines, and changelog generation — enabling scriptable cluster, job, and notebook management for enterprise automation workflows.

- Designed the **REST API** layer and **Java SDK** (published to **Maven Central**), enabling programmatic infrastructure provisioning, job submission, and status polling across all supported cloud providers.
- Designed and shipped the **interactive notebook infrastructure** with real-time coauthoring, version control, integrated data visualisation, and automatic dependency management for PySpark and Spark SQL workloads.
- Built **multi-cloud Terraform modules** (GCP, Azure, AWS) with remote state management (GCS/S3 backends), workspace isolation per tenant, and Sentinel/OPA policy-as-code guardrails for SOC 2 compliance.
- Implemented **tenant isolation architecture** with dedicated VPCs, network policies, RBAC, and namespace-level Kubernetes isolation — supporting **10+ enterprise tenants** with HIPAA/GDPR-compliant data residency controls.
- Managed **Apache Spark runtime versioning** with Long-Term Support (LTS) releases, integrated **data catalog support** (Hive Metastore, Unity Catalog, Delta Lake, Apache Iceberg), and enabled **CUDA GPU acceleration** for ML/AI workloads.
- Reduced deployment cycle time by **40%** (from 45 min to 27 min), by automating end-to-end CI/CD pipelines (build → scan → test → deploy → rollback) via GitHub Actions with Docker multi-stage builds, Helm chart packaging, and automated blue-green and canary releases.
- Owned **product release management** end-to-end — defined release cadences, authored structured release notes, managed version tagging (semantic versioning), and coordinated cross-team release trains across UI, API, CLI, and SDK artefacts.
- Built **automated test pipelines**: integration tests (API contract validation), **OWASP ZAP** security scans (DAST) on every PR, **Selenium-based UI regression** tests, and end-to-end smoke tests — achieving **95%+ test coverage** on critical paths.
- Shipped production SaaS listings across **AWS, Azure, and GCP Marketplaces** with automated licensing, metered billing, entitlement APIs, and subscription management — reducing customer onboarding from **2 weeks to 3 days**.
- Provisioned auto-scaling **Apache Spark clusters** on Kubernetes + Terraform, cutting environment setup by **75%** and infrastructure costs by **50%** (~\$70K/month saved).
- Evaluated and integrated **LXC containers** and **Firecracker microVMs** as lightweight compute alternatives to traditional VMs, reducing **VM bootstrap time from 3–5 minutes to under 30 seconds** and improving cluster cold-start performance for ephemeral Spark workloads.
- Designed **cost-aware billing dashboards** with real-time cloud spend visibility per tenant, automated budget alerting, and resource tagging enforcement across all three cloud providers.
- Built a **no-code migration utility** to ingest Spark jobs, notebooks, and configurations from Databricks, Dataproc, and EMR, converting them into platform-ready pipelines — reducing migration effort by **80%**.
- Implemented **comprehensive audit logging** and **RBAC with fine-grained access control** across all platform operations (cluster, job, notebook, tenant), enabling SOC 2 and HIPAA compliance audit trails.
- **Led pre-sales engineering and product pitching** — delivered live product demos to Fortune 500 prospects, designed and executed **enterprise POCs** end-to-end, and participated in technical RFP responses that secured new contracts.
- Showcased the platform at the **Data Engineering Summit (DES)** organised by AIM, India's premier data engineering conference — presenting architecture deep-dives and live demos to 500+ attendees.
- Represented engineering in **15+ executive briefings, enterprise panels, and customer advisory calls**, directly influencing **multi-million-dollar contract renewals** and shaping product direction for key Fortune 500 accounts.

Senior DevOps Engineer | Modak Analytics, Hyderabad

Mar 2022 – Aug 2024

Embedded Platform Engineer for a life-sciences data platform — responsible for infrastructure architecture, tooling standardisation, and cloud cost governance across 5 product teams.

- Served as **embedded Platform Engineer**, leading technical stack evaluation, tooling standardisation, and infrastructure architecture decisions across the engineering org.
- Built **Terraform golden modules** for VPC provisioning (peering, shared VPC, private service access), JFrog Artifactory, and Splunk integration — reducing new-tenant onboarding from **3 days to under 4 hours**.
- Authored **reusable Helm charts** for all internal microservices with standardised health checks, resource limits, liveness/readiness probes, and Prometheus ServiceMonitor annotations.
- Reduced deployment errors by **70%** and release cycle time by **50%** via GitOps-driven CI/CD (GitHub Actions + Azure DevOps) with automated linting, unit tests, integration tests, container scanning (Trivy), and SAST (SonarQube).
- Architected **on-demand compute orchestration** across GCP Preemptible VMs, AWS Spot Instances, and Azure Spot VMs with automated failover, state checkpointing, and graceful drain — achieving **zero data loss** on spot interruptions and **40% compute cost savings**.
- Implemented **Kubernetes cluster management** at scale: HPA (Horizontal Pod Autoscaler), PDB (Pod Disruption Budgets), RBAC, Network Policies, resource quotas, and namespace isolation across multi-tenant environments.
- Hardened multi-cloud security posture by implementing **ingress/egress firewall rules**, VPC Service Controls, security groups, and **shift-left vulnerability scanning** (JFrog Xray + Snyk) in CI/CD — achieving **zero critical CVEs** in production for 12+ months.
- Designed **cloud cost governance framework**: automated billing dashboards (Grafana + BigQuery/Azure Cost Manage-

ment), tagging enforcement via Terraform, and budget alerting with PagerDuty escalation — achieving **25–35% monthly spend reduction** (~\$25K/month saved).

- Built a **real-time observability stack** (Grafana, Prometheus, Alertmanager, ELK, Splunk, Stackdriver) with standardised dashboards, SLI/SLO-driven alerting rules, and incident runbooks — reducing **MTTR from 4 hours to 2.4 hours (40%)**.
- Integrated **Apache Spark History Server** and Spark UI proxying into the platform observability layer for job-level performance monitoring, stage analysis, and executor memory profiling.
- Designed **automated infrastructure drift detection** using Terraform plan diffs in CI/CD, with Slack/PagerDuty notifications and auto-remediation for common configuration drift patterns.
- Built **self-service developer portals** enabling product teams to provision cloud resources, spin up environments, and manage secrets (HashiCorp Vault, GCP Secret Manager) without DevOps intervention.
- Deployed **Confluent Kafka** (Schema Registry, Connect, ksqlDB) for real-time event streaming between platform services, handling **50K+ events/minute** with exactly-once delivery guarantees.
- Delivered **marketplace-ready SaaS integrations** (AWS, Azure, GCP) with automated licensing, usage metering, billing pipelines, and entitlement verification via cloud-native APIs.

DevOps Engineer | Modak Analytics, Hyderabad

Apr 2020 – Feb 2022

Infrastructure engineer for a life-sciences data platform — customer-facing cloud deployments across GCP and Azure, with a focus on networking, database management, and containerisation.

- Provisioned **customer-specific infrastructure** on GCP and Azure (Compute Engine, Cloud Run, PostgreSQL, GCS, ADLS Gen2, VPCs, subnet design, firewall rules) via parameterised Terraform modules, reducing setup time by **60%**.
- Managed **MongoDB Atlas** clusters on GCP and Azure with geo-based, hash-based, and range-based sharding strategies, achieving **3× query performance improvement** and multi-region data residency compliance (GDPR, HIPAA).
- Enabled **SSO via Azure AD** (SAML 2.0, OAuth 2.0) for internal tooling and customer-facing SaaS applications across **200+ users**, reducing password-reset tickets by **80%** and achieving SOC 2-compliant identity management.
- Built **multi-tenant VPC topologies** on GCP with VPC peering, shared VPC, private Google access, and dedicated subnets per tenant — achieving SOC 2-compliant network isolation for **8 enterprise customers**.
- Integrated **JFrog Artifactory** as a universal artifact repository (Docker, Maven, PyPI, Helm) with Xray vulnerability scanning, ensuring all production images pass security gates before deployment.
- Deployed **containerised microservices** on Kubernetes with rolling updates, health checks, pod disruption budgets, and multi-region failover — achieving **99.95% uptime** with zero-downtime deployments.
- Automated **StreamSets Data Collector** and **Confluent Kafka** deployments across GCP and Azure via Ansible playbooks and Terraform, reducing pipeline failure rates by **60%**.
- Achieved **sub-2-hour bottleneck resolution** by implementing full-stack observability via Splunk (log aggregation, custom dashboards, alerting) and Dynatrace (APM, distributed tracing, code-level profiling).
- Designed **backup and disaster recovery** strategies for PostgreSQL and MongoDB with automated snapshots, cross-region replication, and point-in-time recovery — achieving **RPO < 1 hour** and **RTO < 2 hours**.

Data Engineer | Modak Analytics, Hyderabad

Sep 2019 – Mar 2020

Built enterprise data pipelines processing 1TB+ daily across multi-cloud environments using Java, StreamSets, and Apache Kafka.

- Built a **Java-based ETL platform** on StreamSets Data Collector + PostgreSQL, handling **1TB+ daily data ingestion** across GCP, Azure, and AWS for 5 enterprise data sources (Oracle, MongoDB, S3, ADLS, GCS).
- Engineered **reusable pipeline connectors** for Apache Kafka, GCS, S3, ADLS, Oracle, and MongoDB with custom error-handling, retry logic, and dead-letter queues — reducing integration effort for new data sources by **70%**.
- Automated **end-to-end PaaS provisioning** (Cloud SQL, Cloud Run, GCS buckets, Azure App Service, ADLS containers) across three clouds via Terraform and Bash, improving deployment speed by **60%**.
- Implemented **data quality validation** pipelines with schema drift detection, null-check assertions, and row-count reconciliation, ensuring **99.5% data accuracy** across ingestion workflows.

PROJECTS & OPEN SOURCE

Terraform Automator — GitHub · HCL, GitHub Actions, GCP

- Built a **GitHub Actions-driven Terraform pipeline** for GCP with automated plan, apply, and destroy stages, approval gates, remote state locking (GCS backend), and Slack notifications on drift detection.

Docker Watcher — GitHub · Python, Flask, Docker API

- Developed a **real-time Docker container monitoring** web app displaying CPU, memory, network metrics, and health status with WebSocket-based live refresh and colour-coded status badges.

EFMK Monitoring Stack — GitHub · EFK, Kafka, Docker Compose

- Assembled a production-grade **Elasticsearch + Fluentd + Kibana + Kafka** observability stack for centralised log

aggregation, real-time alerting, and dashboard visualisation across containerised microservices.

Nginx Multi-Service Proxy — [GitHub](#) · Nginx, Docker, Reverse Proxy

- Built a **Docker Compose–based multi-service architecture** with Nginx reverse proxy, demonstrating service routing, load balancing, SSL termination, and upstream health checking.

Donor Near Me — [GitHub](#) · Flutter, Python, Firebase

- Built a **Flutter PWA** with a Python/Flask backend for real-time blood-donor discovery featuring geolocation, push notifications, donation cooldown tracking, and donor certificate generation.

EDUCATION

B.E. Electronics & Communication — Osmania University, Hyderabad

Jun 2019